

Ayush Pandey

B.Tech Mathematics & Computing · RGIPT, Amethi, India · CGPA 8.68/10

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RESEARCH PROFILE

First-year undergraduate researcher in Mathematics and Computing with hands-on experience in statistical physics, probabilistic computing, and ML for physical systems. Currently contributing to an active Quantum Materials Lab and building open-source tools at the intersection of physics-informed ML and systems programming.

EDUCATION

B.Tech in Mathematics and Computing

2025 – 2029

Rajiv Gandhi Institute of Petroleum Technology (RGIPT), Amethi, India

Relevant Coursework: Linear Algebra · Probability Theory · Differential Equations · Modern Physics · C Programming · MATLAB

RESEARCH EXPERIENCE

Research Intern – Muon Spin Relaxation (μ SR)

Jan 2025 – Present

Quantum Materials Lab, RGIPT · sites.google.com/rgipt.ac.in/magneism-lab

- Apply supervised ML to predict muon stopping sites in crystalline materials, integrating physics constraints directly into model architecture.
- Build end-to-end data pipelines for analyzing spin relaxation spectra using *pymatgen* and domain-specific experimental datasets.
- Bridge condensed matter theory with data-driven modelling — relevant to scientific ML research in quantum materials.

PROJECTS

Rust-Based Neural Network Framework

2025 – Present

github.com/TheOrganic-code · Systems ML · Ongoing

- Building a neural network framework from scratch in Rust, focusing on memory safety, performance, and zero-cost abstractions.
- Implementing core components: tensor operations, automatic differentiation, and modular layer APIs without external ML dependencies.
- Targeting high-performance inference pipelines where Python-based frameworks incur prohibitive overhead.

Physics-Informed Probabilistic Computing (p-bit) Simulator

2025 – Present

github.com/TheOrganic-code/P-bit-Simulator · Stochastic Computing · Energy-Based ML

- Built stochastic computing simulator grounded in Ising spin systems and Boltzmann machines with tunable thermal noise parameters.
- Modelled energy landscapes; verified convergence via energy minimisation on MAX-CUT benchmarks — core energy-based ML methodology.
- Designing reusable p-bit library with PyTorch-compatible APIs for hybrid deterministic–stochastic workflows (ongoing).

Brain Tumor MRI Classifier with Grad-CAM Interpretability

2025

github.com/TheOrganic-code/Brain-Tumor-Classifer · Deep Learning · Explainable AI

- CNN-based MRI tumour classifier in PyTorch; applied Grad-CAM for attention visualisation and diagnostic interpretability.
- Focused on model trustworthiness and explainability — directly relevant to responsible AI and ML safety research.

Planetary Habitability Predictor

2025

github.com/TheOrganic-code/Habitability-Predictor · Scientific ML

- Random Forest and Logistic Regression on planetary datasets; feature importance analysis on atmospheric and orbital parameters.

TECHNICAL SKILLS

Languages: Python · C · C++ · Rust

ML / Deep Learning: PyTorch · scikit-learn · CNNs · Energy-Based Models · Grad-CAM

Data & Scientific: NumPy · Pandas · pymatgen · Matplotlib · Seaborn · MATLAB

Systems & Tools: Git · GitHub · Linux · Rust (systems programming)

Mathematics: Linear Algebra · Probability Theory · Optimisation · Differential Equations

RESEARCH INTERESTS

ML for Physical Systems · Energy-Based Models · Probabilistic Computing · Scientific ML · Quantum Materials · Deep Learning Interpretability · Systems ML · Quantum Computing

HIGHLIGHTS

- First-year undergraduate actively contributing to peer-level research at Quantum Materials Lab, RGIPT.
- Building a Rust-based neural network framework from scratch — demonstrating rare systems-level ML depth.
- Open-source contributor: p-bit simulator and ML projects publicly available on GitHub with documented methodology.
- Self-driven learner: independently acquired PyTorch, Rust, stochastic modelling, and pymatgen skills alongside coursework.